Installation Instructions
AFL Optical Ground Wire (OPGW)
(Stainless Steel Tube & Aluminium Pipe Cable Designs)

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1.0 General Information

Composite Optical Ground Wire (OPGW) was developed to provide a large capacity telecommunications system utilizing overhead power transmission lines. Serving the additional purpose of an overhead ground wire, the OPGW is constructed of aluminum clad steel strands and aluminum alloy strands stranded with stainless steel tubes or surrounding a fibre unit (core) which contains optical fibres. OPGW can be installed using the basic stringing methods currently employed for overhead ground wires, with minor variations.

This document outlines basic installation methods applicable for existing and newly constructed transmission lines. The installer should be thoroughly familiar with the installation of conventional overhead ground wire and conductors. Additional information can be obtained from the latest revision of IEEE Guide to Installation of Overhead Transmission Line Conductors, IEEE Std 524.

2.0 Precautions

Care must be taken to avoid damaging the OPGW during handling and stringing operations. Avoid sharp bends to the cable and take precautions to prevent crushing the OPGW during placement. The transmission quality of the optical fibres can potentially be degraded if the OPGW is subjected to excessive pulling tensions or excessively small bend diameters.

Always observe the recommended values for Maximum Stringing Tensions and Minimum Bend Radius. More information about these values is contained on the following pages.

OPGW is normally supplied on non-returnable wooden reels. The cable is packaged with a flex wrap or wooden lagging to provide additional protection during transportation. If the cable is not to be installed for a period of over four months from the delivery date, it is recommended that the cable be requested and shipped on steel reels rather than wood reels. Please contact AFL for more details or to request shipment on steel reels.

OPGW cable reels should always be transported and handled in an upright position. Never lay a reel of cable on its side. It is recommended that each reel of OPGW be tested prior to and after installation to ensure that fibre damage has not occurred during shipping and/or stringing operations. All cable protective packaging (wood lagging or flex wrap) must remain in place on all reels until placed on a pay-out rack and the rack is in position for cable stringing.

Above all, be familiar with and observe all of your company’s safety rules when working with overhead transmission lines. These installation recommendations should not supersede any established safety practices.

3.0 Cable Installation

Reel Preparation Prior to Beginning a Pull

AFL ships the cable reels with the inner tail securely connected to the outside of the reel flange. This connection should be loosened, but not removed, prior to stringing. This allows the inner layers of cable to adjust themselves to the varying tensions seen during installation. As the cable makes these adjustments, the inner tail may lengthen, or “grow,” requiring periodic attention to ensure that the cable continues to be in a state where it can “grow” out.

The wooden reel will have through bolts connecting the two flanges. During shipment, these bolts can loosen. Prior to stringing, the bolts should be tightened to help prevent any issues while paying off the cable.

AFL recommends using the controlled tension stringing method of installation. Ordinary stringing equipment can be utilized as if installing standard overhead ground wire provided all of the minimum block sizes and other requirements of these instructions are followed. Suitable equipment includes pullers, tensioners, reel winders, and stringing blocks.

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Figure 1 illustrates a typical stringing setup

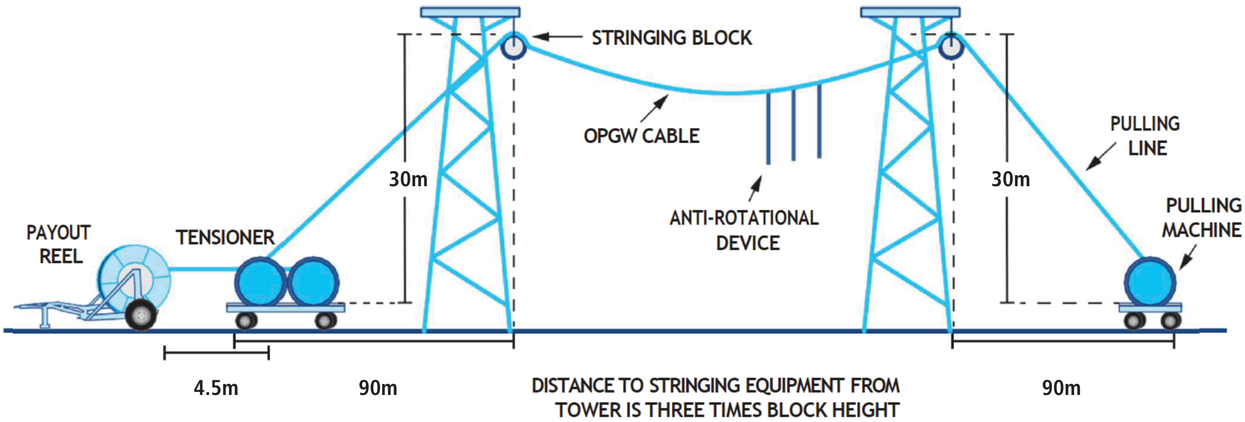

Figure 1 - Typical Stringing Setup

There is one primary difference between installing OPGW and conventional overhead ground wires. Standard ground wires are typically spliced using compression type connectors and locations of the splices are relatively flexible. The splice locations of the OPGW cable must be planned to allow for splicing of the optical fibres. The reel lengths will be engineered to locate the cable splices at predetermined towers on each end of a stringing section.

After installing dead ends, the free ends of the OPGW are trained down the towers to an additional 23 meters to accommodate the splicing. After stringing, this cable length is typically coiled and temporarily stored at the tower until the splicing occurs.

The OPGW will also use special attachment hardware, including dead ends, suspension clamps, and wire fittings such as grounding clamps. The hardware is designed to provide the necessary holding strengths and prevent deformation of the fiber unit which could potentially damage the optical fibres.

Table 1 - Temperature Ranges

Storage	-50ºC to +85ºC
Installation	-30ºC to +85ºC
Operation	-40ºC to +85ºC

4.0 Stringing Procedures

Stranded wire pulling lines are generally used, although nylon ropes have also been employed. In either case, the line must be rated strong enough to withstand the required stringing tensions. The pulling line should have the same direction lay as the OPGW to help resist the tendency to rotate under stringing load.

If an existing overhead ground wire is to be removed, it can potentially be used as a pulling line for the OPGW. A visual inspection should be made of the existing ground wire to be sure it is in suitable condition. If there is any concern about the existing wire's ability to withstand the stringing tensions, it should be pulled out and replaced with a pulling line.

It is recommended to use a bull wheel type tensioner with round (not 'V' type) polyurethane lined contact grooves. The tensioner should have two bull wheels, each with multiple grooves to minimize cable damage. The tensioner should be capable of maintaining the required tensions at various pulling speeds. Positive braking systems are necessary for pullers and tensioners to maintain the tension when pulling is stopped. Minimum diameter of the bull wheels should not be less that 70 x D (diameter of the OPGW). For cable diameters greater than 20 mm, please contact AFL.

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The OPGW must be reeved (threaded) through the bull wheel tensioner properly. Left hand lay OPGW (typical USA) is reeved from right to left, as shown in Figure 2. Right hand lay OPGW (typical International), is reeved from left to right. A thorough explanation of the reeving process can be found in the latest revision of IEEE Std 524. This arrangement is necessary to avoid any tendency to loosen the outer layer of strands and to avoid induced torque during installation.

The reel shall be placed directly in line with the tensioner. The distance from the reel to the tensioner should be at least 7.5 meters. The OPGW shall not be permitted to scrape the reel flanges while being pulled.

The OPGW cable reels are not designed to withstand the braking forces present during stringing. Direct tensioning of the OPGW from the cable reel is not recommended. Back tension on the reel should only be enough to keep the cable properly seated in the tensioner grooves and to prevent overshooting and bird-caging.

Figure 2 - Figure 2 - Reeving direction for Left Hand Lay OPGW

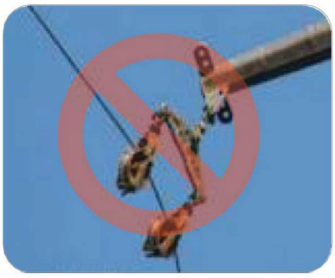
Two basic types of pulling machines are recommended for tension stringing. These are either drum type or bull wheel type pullers. Positive braking systems are required in either case. On a drum type puller, the pulling line is taken up directly onto the drum. On the bull wheel type, the line is threaded onto two bull wheels, much like the tensioner, and onto a self winding drum.

Stringing blocks, sometimes called travelers or sheaves, are mounted on the structure at the OPGW attaching point in the normal manner. Please refer to Reference A, OPGW Diameters and Bending Radius, for information on blocks diameters.

The stringing blocks should have neoprene lined grooves. The linings should be in good condition and adhering to the block. Minor rough areas can be sanded out to ensure the lining is smooth.

'Radius Blocks,' 'Banana Blocks' or 'Array Travelers' as shown below in Figures 3A-C are strictly prohibited during the installation of OPGW. The decreased surface area in contact with the OPGW is sufficient to damage the OPGW at typical stringing tensions.

A



B

C

Figures 3A-C - These types of array installation blocks should NOT be used to string OPGW

Uplift rollers (which attach to the installation sheave wheel) or hold-down blocks (which are separate blocks) need to be placed where uplift of the pulling line is likely to occur (due to its higher tension/weight ratio than the conductor). This will typically occur going up inclines or at a low point in a section. These devices should also have a break away feature in the event of fouling or incorrect installation. The size of the uplift rollers should follow the same guidelines as the installation sheaves shown in Reference A.

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The tensioner and puller should be positioned for a 3:1 ratio to the stringing block on the first structure adjacent to the equipment. See Figure 1. The tensioner should be placed in line with the first two structures (or first span) of the pull. Likewise, the puller should be placed in line with the last two structures (or last span) of the pull. Doing so minimizes the line angle change seen by the cable during the installation process.

This minimum stringing block diameter and distance to the tensioner (3:1) are recommended to help prevent deformation of the fibre unit (aluminum pipe, stainless steel tube or slotted core), which protect the optical fibres in the OPGW.

The use of an **Anti-Rotational Device** (see sample drawings on pages 14 and 15) depends largely upon the construction of the optical ground wire. Such a device is used to prevent the OPGW from twisting while being pulled. Variations of these devices have been successfully used. Please consult AFL for any inquiries regarding a particular form of anti-rotation device.

For cables with helically stranded stainless steel tubes, an anti- rotational device may or may not be required. To confirm whether one is needed for your particular application, contact AFL. When in doubt, the conservative approach is to conduct the installation with the use of an anti-rotational device. For cables constructed with an un-stranded stainless steel tube in the center of the cable or single layer cables, an **anti-rotational device is always a requirement**. (See Appendix 1 on page 14 and Appendix 2 on page 15.)

If the anti-rotational device is not preventing the cable rotation or if the anti-rotational device is wrapping around the OPGW, a stiffer or heavier device is required. The weight and length of the ARD will depend upon the construction of the optical ground wire.

The anti-rotational device attaches to the OPGW with a Kellum type grip. The grip must be appropriately sized for the OPGW diameter and pulling tensions.

Normally, the OPGW should be kept under constant stringing tension during the stringing process to keep the line clear of both the ground and other obstacles that could cause damage to the cable.

Do not cut the OPGW with ratchet cutters or other types of tools that could crush or crimp the optical core. The use of a hacksaw will ensure the fibre optic units are free to move within the pipe. During stringing, the optical fibre core (design dependent) may pull back into the cable, requiring a few feet to be cut away upon splicing on the leading end to expose the optical core.

It is important to monitor the tensions and ensure that excessive tension is not applied as the OPGW passes from the reel to the tensioner.

Table 2 on the following page show the recommended values for safe OPGW installation.

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Table 2 - Recommended Values for Safe OPGW Installation

PARAMETER	VALUE	
Minimum Bull Wheel Diameter	70 x OPGW diameter	
	For larger diameter OPGW cables where 70 x OD exceeds ~1.5 m, a ~1.5 m bull wheel may be used. Please consult AFL in such cases.	
Recommended Block Diameter for First and Last Structure	40 x D	
	Smaller diameters can be used at tangent structures. See Reference A.	
Minimum Cable Bend Radius	During Installation (Dynamic):	20 x OPGW diameter
	After Installation (Static):	15 x OPGW diameter
Maximum Stringing Tension	20% of the Rated Breaking Strength of OPGW	
	The stringing tension is always measured at the tensioner side. In general, the maximum stringing tension should be a half of the maximum sagging tension and never should exceed 20% RBS of the OPGW.	
Pulling Speed	60 meters per minute **OR** 3.6 km per hour	
Minimum Distance from Puller and Tensioner to the Stringing Block	3:1 Ratio	
Total Number of Spans in each Stringing Section	Typically 20 to 30*	
	* The maximum number of spans is included as a reference only; since this will vary considerably due to differences in terrain, span lengths, line angles, etc.	

5.0 Sagging Methods

The methods and procedures for sagging the OPGW are the same as those for normal overhead ground wires. For determining sags, the installer should use the sag-tension design information provided by the utility or AFL. Sagging and tensioning should be conducted from dead end to dead end. Care should be taken to avoid sagging the cable around angles greater than 30 degrees.

A temporary grip is installed on the OPGW for tensioning. The grip must be designed to aluminum pipe. AFL can provide a come-along, sometimes called a pocketbook grip, that can be attached anywhere along the length of an OPGW. Figure 4 illustrates a satisfactory come-along design.

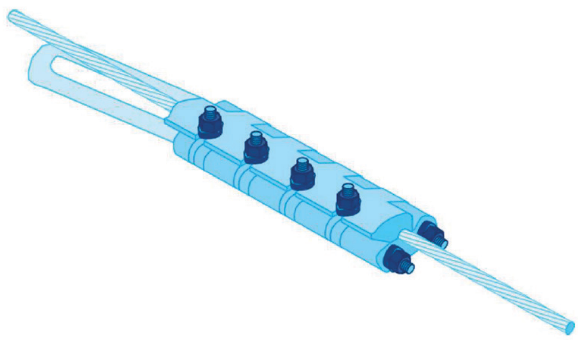

Figure 4 - Come-along (Pocketbook Grip)

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Some types of gripping devices that might induce crushing damage to the OPGW such as Chicago grip or Kito grip are strictly prohibited to use for OPGW, as shown in Figures 5A and B.

Kito Grip (prohibited)

Chicago Grip (prohibited)

Figures 5A and 5B - Types of prohibited grips used to tension OPGW.

Certain types of formed guy grips can also be used successfully, but their use in stringing applications should be checked with the grip's manufacturer.

6.0 Dead Ending and Clipping In

Dead Ends are installed on OPGW spans that terminate at splicing towers or ends of the system. Dead Ends are also used at angle structures when the angles are too great to use suspension clamps. Suspension clamps are normally used at the remaining towers. These types of hardware (dead end and single suspension) are illustrated in Figures 6 and 7.

Figure 6 - OPGW Dead End (Bolted Type)

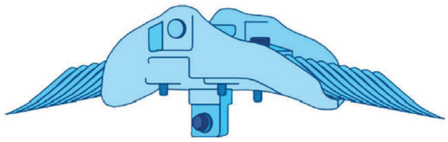

Figure 7 - OPGW Suspension Unit

In general, the rule for hardware use is the following:

- Single Suspension to be used at structures with line angles between 0 and 30 degrees.
- Double Suspension to be used at structures with line angles between 30 and 60 degrees.
- Double Dead End to be used at structures with line angles over 60 degrees.

Dead ends can be used starting from line angles of 30 degrees in lieu of double suspensions.

In order to diminish the probability of motion-induced damages and creep rate change, AFL recommends that tensioning and anchoring of the OPGW to the structure and removal of the stringing blocks be completed no later than 48 hours after pulling in the cable.

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OPGW is installed using stringing blocks. If left in the stringing blocks for extended periods of time, the potential for motion induced damage (aeolian vibration) increases. Also, the creep of the cable is affected due to the change in the initial condition on the cable. In order to diminish the probability of motion induced damages and creep rate change, AFL recommends that tensioning and anchoring of the OPGW to the structure and removal of the stringing blocks be completed no later than 48 hours after pulling in the cable.

There are several ways to lift the OPGW from the stringing blocks in order to install the hardware. A come-along is attached on both sides of the block and a coffin hoist is placed over the tower arm. The hooks of the coffin hoist are attached to the come-along and jacked up to form a small loop in the OPGW. The block can then be removed and the armor rods can be placed on the OPGW then attached to the structure. Alternately, certain types of preformed wire grips can be used instead of come-along. The preformed grip can be used once as a come-along and then used permanently in the next span.

If vibration dampers are required for this span, these should be placed on the OPGW immediately after clipping in. Dampers may not be required at every structure; their locations will be specified by the utility or AFL. A drawing of an AFL Stockbridge damper is shown in Figure 8.

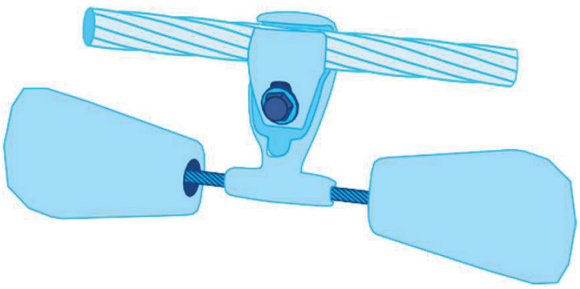

Figure 8 - Stockbridge Damper

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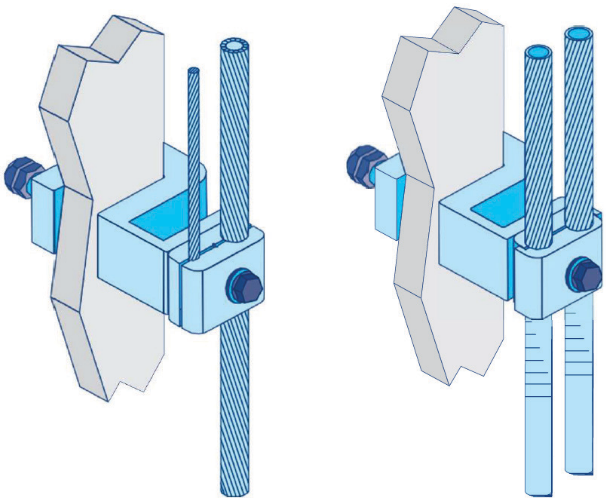
7.0 Splice Points

Splice points will be located at the beginning and end of each OPGW reel. After completion of sagging and clipping, the surplus OPGW should be coiled and attached temporarily to the tower. Coils should be approximately 1 to 1.5 meters in diameter. The coils should be fixed on the tower to prevent any damage to the OPGW prior to splicing.

The exposed ends of the OPGW should be re-sealed to prevent moisture from entering the fiber units. The cable reel may be supplied with a pair of plastic caps for sealing the cable ends. Electrical tape, RTV silicone, or other means can also be used for this purpose.

The OPGW will be trained down the tower and to the ground for splicing. Do not cut off any excess length of the OPGW at this time. To facilitate splicing, the OPGW should extend beyond the bottom of the tower. The length of OPGW running down the tower should be attached to the structure using appropriate guide clamps, spaced every 1.8 to 2.4 meters of running length. Several types of guide clamps are illustrated in Figure 9.



Tower Bonding Clamp **Tower Guide Clamp**

Figure 9 - OPGW Guide Clamps and Bonding Clamps

The splice enclosure will typically be installed on the structure between 4.5 to 6 meters above the ground. In most cases, it will be desirable to store extra cable on the tower. This will allow the splice box to be removed and lowered to the ground if it is ever necessary. This can be accomplished with a simple loop of OPGW below the splice box or by permanently storing a coil of OPGW higher on the tower.

After the stringing procedure, the ends of the optical unit(s) must be located. It is recommended that this occurs prior to the cable being cut free from the payoff to ensure proper length remains for splicing. In some OPGW designs there could be some pull back of the optical unit(s) during the stringing installation. 'Pullback' is a term used when the optical unit(s) or core appears to migrate inside the pipe due to elongation of metallic components during the stringing procedure. When this happens, the core must be located so the proper amount of optical unit(s) is available for the installation. Cut back 1 to 1.5m at a time until the optical unit(s) is found.

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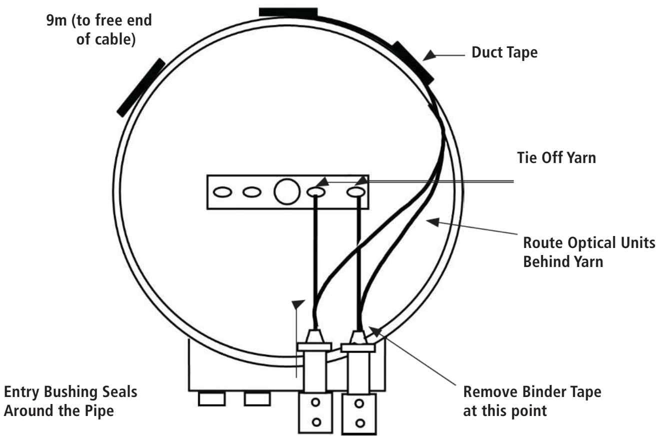
8.0 Anchoring the Optical Units in the Splice Enclosure

The following is an example of anchoring the optical units when installing AFL Loose Tube OPGW (Aluminum pipe designs) in an AFL SB01-72 Splice Enclosure (See **Figure 10**).

- Once the pipe is removed, the fibre core should be cut to length and the binders removed.
- Unwind the optical units and set aside the aramid yarn in the center of the unit.
- Wind the optical units back into their original position and use electrical tape every 40cm to secure the units. Leave the end of the units free of tape so that the units can be prepped for splicing.
- Remove any excess aramid yarn, leaving at least 35 - 45cm to tie off into the box.
- Secure the yarn to the eye bolt in the enclosure. (See Figure 10)
- Tape the units (duct tape is recommended) to the top outside radius of the box (See **Figure 10**). This will help support the units and prevent damage of the units at the bushing.
- Tighten the retaining nut of the connector kit so that the bushing is sealed around the pipe.
- Repeat the previous steps for the remaining OPGW cable(s).

A similar procedure is available for AFL Tight Structure, HexaCore and CentraCore OPGW and AFL Opti-Guard Splice Enclosure.

** Ensuring that the optical core is 'tied off' within the chosen splice enclosure (whether AFL splice enclosures or not) is a must. If not, the optical core may work its way back into the cable over time, damaging the splice box contents and potentially affecting the optical continuity.**



NOTE: There should be 9m of optical unit (s) prepped for each individual OPGW cable that is to be inserted into the Splice Box. The 9m of optical unit (s) is divided two sections:

1: 6m from box to ground, and

2: 3m for storing and splicing. Should the Splice Box be mounted at a different height, then adjust the amount of OPGW cable to be prepped, stored and spliced.

Figure 10 - Anchoring the Optical Units in the AFL SB01 Splice Enclosure

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REFERENCE A

Stringing and Handling OPGW Diameters and Bending Radius

The following guides apply to AFL OPGW Cable.

Table 3 - Recommended Values for Stringing and Handling OPGW

PARAMETER	VALUE	
Maximum OPGW Stringing Tension	20% (at tensioner) of the rated breaking strength	
Minimum Bull Wheel Diameter	70 x OD x OPGW	
	For larger diameter OPGW cables where 70 x OD exceeds ~1.5 m, a ~1.5 m bull wheel may be used. Please consult AFL in such cases.	
Stringing Sheave (Root) Diameter	40 x OD of OPGW	
	Based on a sheave through angle of 45° and maximum stringing tension (at tensioner) of 20% of the rated strength of the OPGW.	
	NOTE: Refer to Table 4 for additional information on minimum diameters of the stringing blocks for other conditions.	
Minimum Cable Bend Radius	After Installation (Static):	15 x OD of OPGW
	During Installation (Dynamic):	20 x OD of OPGW
	NOTE: Based on actual OPGW size, etc., care must be taken when bending the OPGW to avoid kinking the strands and damaging the optical fibers contained within the central pipe.	
Minimum Permanent Bending Radius	OPGW:	15 x OD of OPGW
	Stainless Steel Tube:	45 x OD of stainless steel tube
	Plastic Buffer Tubes:	8 cm
	Optical Fibers:	3.8 cm
Swinging Angle of Stringing Block	Shall be controlled corresponding to the swinging angle of the OPGW stringing plane to help prevent the cable from riding out of the traveler or excessive twisting during installation.	
	The cable should travel through the lowest part of the groove.	
	At angle structures, this is done by tying a support rope to the sheave to keep it suspended as shown in Figure 11.	

Figure 11 - Tying to support rope to the sheave keeps it suspended to the stringing block.

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The following are minimum diameters of stringing blocks at:

Sheave Size Recommendations

The following sheave diameters are recommended for their respective stringing angles in **Table 4**. These sizes are considered satisfactory if the pulling line slope is at least three horizontal to one vertical from the traveler to the site and the stringing tension does not exceed 20% of the OPGW's rated breaking strength.

Table 4 - Sheave Size Recommendations Based on Stringing Angle

STRINGING OR LINE ANGLE		SHEAVE SIZE (x OPGW OD)	
Bull Wheel Diameter		70x	
First and Last Structures		40x	
Max. Stringing Tension for Recommendation		20% RBS	10% RBS
Tangent Structure Stringing Angles	0° to 20°	25x	20x
	21° to 55°	40x	30x
	56° to 90°	55x	45x
	≥ 91°	No Go	

NOTE: 45cm Minimum Sheave Size

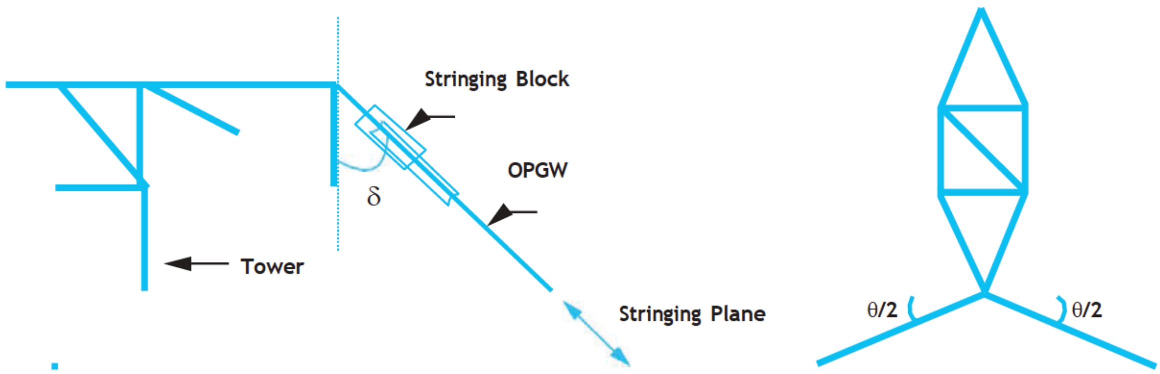

Figure 12 - Stringing angles and sheave size diagram

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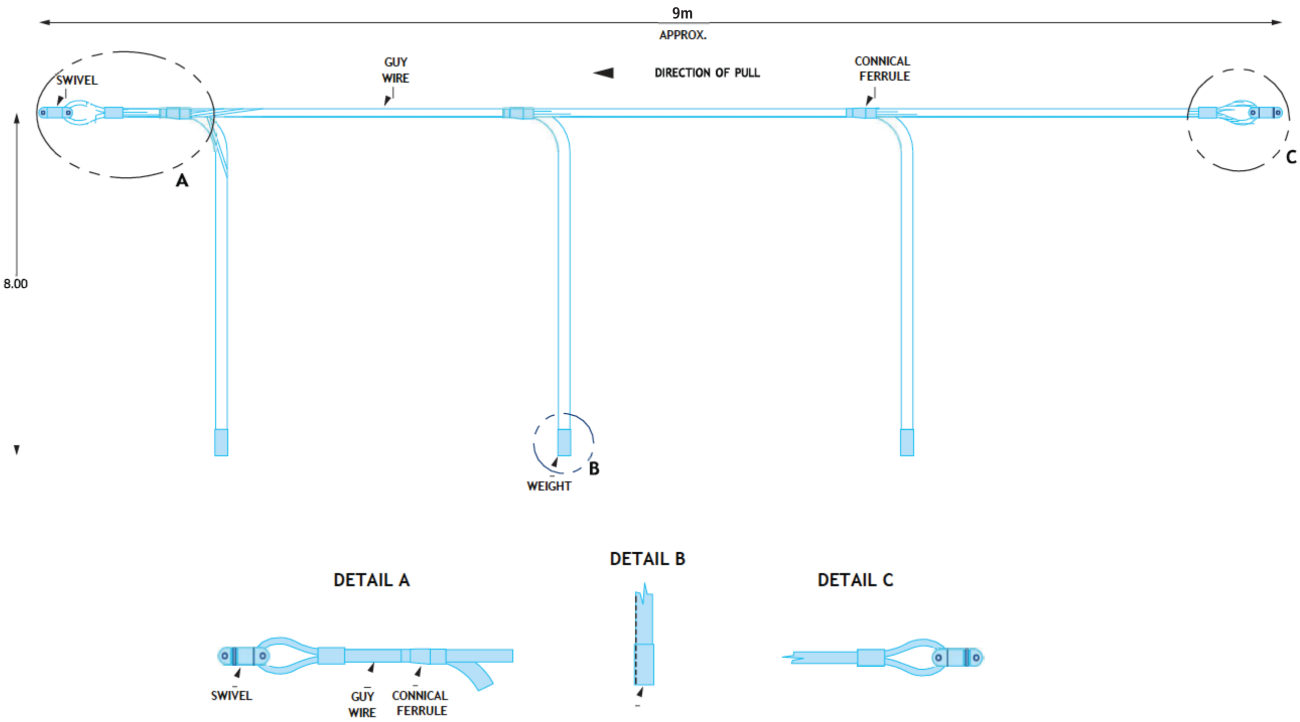
APPENDIX 1 Anti-Rotational Device Example (Attached in Front of the OPGW)



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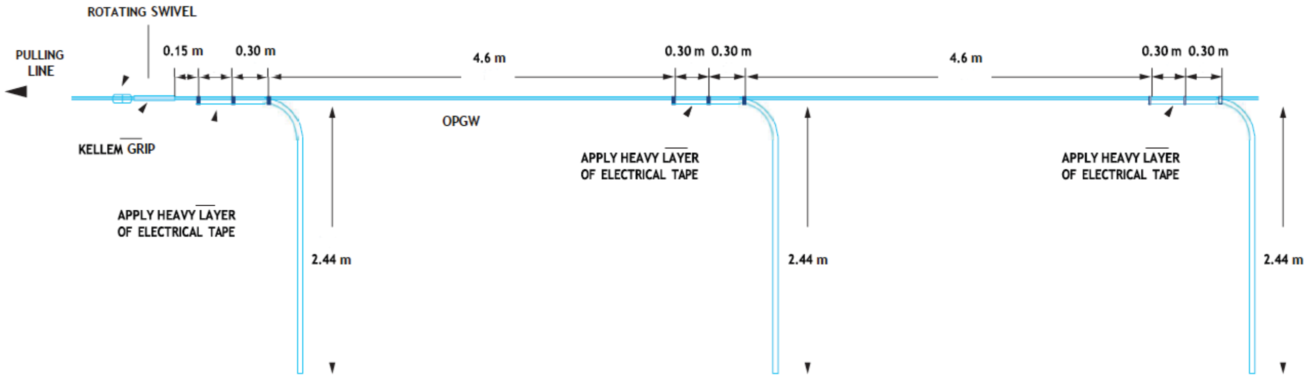
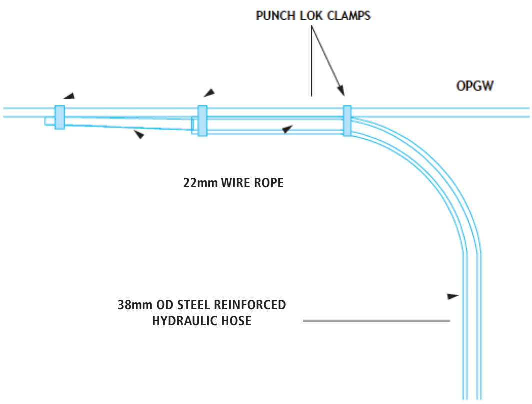
APPENDIX 2 Anti-Rotational Device Example (Attached Directly to the OPGW)





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APPENDIX 3 Sheave Size Based on Line Angle and Installation Tension